

3D Gaussian Splatting – Full Installation, Running, and Troubleshooting Guide (Ubuntu)

This document provides a complete, battle-tested setup guide for running GraphDECO's 3D Gaussian Splatting on Ubuntu Linux. It includes installation procedure, fixes for CUDA and GCC mismatches, COLMAP errors, rendering, metrics, and SIBR viewer instructions.

1. System Requirements

Hardware: NVIDIA GPU (Compute Capability ≥ 7.0), 4 GB VRAM minimum.

Software: Ubuntu 18.04–22.04, CUDA Toolkit 11.6/11.8, Python 3.7, COLMAP, OpenGL 4.5+.

2. Clone the Repository

```
git clone https://github.com/graphdeco-inria/gaussian-splatting --recursive
```

3. Create Conda Environment

Install PyTorch cu116 and CUDA toolkit 11.6.

Ensure nvcc reports CUDA 11.6, not 12.x.

4. GCC/G++ Mismatch Fix

CUDA 11.x supports gcc-11 max. Install gcc-11 and export CC/CXX variables.

5. Build CUDA Submodules

```
pip install -e simple-knn diff-gaussian-rasterization fused-ssim .
```

6. Prepare Dataset (COLMAP)

Convert .bin to .txt: `colmap model_converter ...`

7. Run Training

```
python train.py -s datasets/
```

8. Render

```
python render.py -m output/ -s datasets/
```

9. Metrics

```
python metrics.py -m output/
```

10. SIBR Viewers

Install dependencies → force CMake to use conda CUDA → build → run remote or realtime viewer.

Troubleshooting covers CUDA mismatch, GCC errors, dataset structure issues, SIBR PTX errors, and metrics NaN issues.